

Course Material

B.Tech III YEAR - I Semester



Introduction to Quantum Technologies and Applications

Sub Code:23PC0515

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23PC0515	INTRODUCTION TO QUANTUM TECHNOLOGIES AND APPLICATIONS (Qualitative Treatment)	L	T	P	C
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Course Objectives (COBJ):

- Introduce fundamental quantum concepts like superposition and entanglement.
- Understand theoretical structure of qubits and quantum information.
- Explore conceptual challenges in building quantum computers.
- Explain principles of quantum communication and computing.
- Examine real-world applications and the future of quantum technologies.

Course Outcomes (CO):

CO No:	Course Outcome	Bloom's Level	Mapped POs
CO1	Explain core quantum principles in a non-mathematical manner.	L2	PO1, PO2, PO3
CO2	Compare classical and quantum information systems.	L2 & L3	PO1, PO2, PO3, PO4
CO3	Identify theoretical issues in building quantum computers.	L1 & L3	PO1, PO2, PO3, PO4
CO4	Discuss quantum communication and computing concepts.	L3	PO1, PO2, PO3, PO4
CO5	Recognize applications, industry trends, and career paths in quantum technology	L3	PO1, PO2, PO3, PO6, PO7, PO11

Unit 1: Introduction to Quantum Theory and Technologies

The transition from classical to quantum physics, Fundamental principles explained conceptually: Superposition, Entanglement, Uncertainty Principle, Wave-particle duality, Classical vs Quantum mechanics – theoretical comparison, Quantum states and measurement: nature of observation, Overview of quantum systems: electrons, photons, atoms, The concept of quantization: discrete energy levels, Why quantum? Strategic, scientific, and technological significance, A snapshot of quantum technologies: Computing, Communication, and Sensing, National and global quantum missions: India's Quantum Mission, EU, USA, China

Unit 2: Theoretical Structure of Quantum Information Systems

What is a qubit? Conceptual understanding using spin and polarization, Comparison: classical bits vs quantum bits, Quantum systems: trapped ions, superconducting circuits, photons (non-engineering view), Quantum coherence

and decoherence – intuitive explanation, Theoretical concepts: Hilbert spaces, quantum states, operators – only interpreted in abstract, The role of entanglement and non-locality in systems, Quantum information vs classical information: principles and differences, Philosophical implications: randomness, determinism, and observer role

Unit 3: Building a Quantum Computer – Theoretical Challenges and Requirements

What is required to build a quantum computer (conceptual overview), Fragility of quantum systems: decoherence, noise, and control, Conditions for a functional quantum system: Isolation, Error management, Scalability, Stability, Theoretical barriers:

Why maintaining entanglement is difficult, Error correction as a theoretical necessity, Quantum hardware platforms (brief conceptual comparison), Superconducting circuits, trapped ions, Photonics, Vision vs reality: what's working and what remains elusive, the role of quantum software in managing theoretical complexities.

Unit 4: Quantum Communication and Computing – Theoretical Perspective

Quantum vs Classical Information, Basics of Quantum Communication, Quantum Key Distribution (QKD), Role of Entanglement in Communication, The Idea of the Quantum Internet – Secure Global Networking, Introduction to Quantum Computing, Quantum Parallelism (Many States at Once), Classical vs Quantum Gates, Challenges: Decoherence and Error Correction, Real-World Importance and Future Potential

Unit 5: Applications, Use Cases, and the Quantum Future

Real-world application domains: Healthcare (drug discovery), Material science, Logistics and optimization, Quantum sensing and precision timing, Industrial case studies: IBM, Google, Microsoft, Psi Quantum, Ethical, societal, and policy considerations, Challenges to adoption: cost, skills, standardization, Emerging careers in quantum: roles, skillsets, and preparation pathways, Educational and research landscape – India's opportunity in the global quantum race.

Textbooks:

1. Michael A. Nielsen, Isaac L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 10th Anniversary Edition, 2010.
2. Eleanor Rieffel and Wolfgang Polak, *Quantum Computing: A Gentle Introduction*, MIT Press, 2011.
3. Chris Bernhardt, *Quantum Computing for Everyone*, MIT Press, 2019.

Reference Books:

1. David McMahon, *Quantum Computing Explained*, Wiley, 2008.

2. Phillip Kaye, Raymond Laflamme, Michele Mosca, *An Introduction to Quantum Computing*, Oxford University Press, 2007.
3. Scott Aaronson, *Quantum Computing Since Democritus*, Cambridge University Press, 2013.
4. **Alastair I.M. Rae**, *Quantum Physics: A Beginner's Guide*, Oneworld Publications, Revised Edition, 2005.
5. **Eleanor G. Rieffel, Wolfgang H. Polak**, *Quantum Computing: A Gentle Introduction*, MIT Press, 2011.
6. **Leonard Susskind, Art Friedman**, *Quantum Mechanics: The Theoretical Minimum*, Basic Books, 2014.
7. **Bruce Rosenblum, Fred Kuttner**, *Quantum Enigma: Physics Encounters Consciousness*, Oxford University Press, 2nd Edition, 2011.
8. **Giuliano Benenti, Giulio Casati, Giuliano Strini**, *Principles of Quantum Computation and Information, Volume I: Basic Concepts*, World Scientific Publishing, 2004.
9. **K.B. Whaley et al.**, *Quantum Technologies and Industrial Applications: European Roadmap and Strategy Document*, Quantum Flagship, European Commission, 2020.
10. **Department of Science & Technology (DST), Government of India**, *National Mission on Quantum Technologies & Applications – Official Reports and Whitepapers*, MeitY/DST Publications, 2020 onward.

Online Learning Resources:

- [IBM Quantum Experience and Qiskit Tutorials](#)
- [Coursera – Quantum Mechanics and Quantum Computation by UC Berkeley](#)
- [edX – The Quantum Internet and Quantum Computers](#)
- [YouTube – Quantum Computing for the Determined by Michael Nielsen](#)
- [Qiskit Textbook – IBM Quantum](#)

LECTURE PLAN

UNIT I

S.No.	Topic	No. of Periods	Pertaining CO(s)	Taxonomy Level	Mode of Delivery
1	The transition from classical to quantum physics	1	CO1	L2	MD1, MD5
2	Fundamental principles explained conceptually: Superposition, Entanglement	1	CO1	L2	MD1, MD5
3	Uncertainty Principle, Wave-particle duality	1	CO1	L2	MD1, MD5
4	Classical vs Quantum mechanics – theoretical comparison	1	CO1	L2	MD1, MD5
5	Quantum states and measurement: nature of observation	1	CO1	L2	MD1, MD5
6	Overview of quantum systems: electrons, photons, atoms	2	CO1	L2	MD1, MD5
7	The concept of quantization:	1	CO1	L2	MD1, MD5

	discrete energy levels				
8	Why quantum? Strategic, scientific, and technological significance,	1	CO1	L2	MD1, MD5
9	A snapshot of quantum technologies: Computing, Communication, and Sensing	2	CO1	L2	MD1, MD5
10	National and global quantum missions: India's Quantum Mission, EU, USA, China	1	CO1	L2	MD1, MD5

Assessment Components:

1. Unit Test
2. Assignment
3. Course Seminar
4. Course Quiz
5. Case Study

Mode of Delivery:

- MD1: Oral Presentation
MD2: Tutorial
MD3: Seminar
MD4: Hands On
MD5: Videos

Unit 1: Introduction to Quantum Theory and Technologies

The transition from classical to quantum physics, Fundamental principles explained conceptually: Superposition, Entanglement, Uncertainty Principle, Wave-particle duality, Classical vs Quantum mechanics – theoretical comparison, Quantum states and measurement: nature of observation, Overview of quantum systems: electrons, photons, atoms, The concept of quantization: discrete energy levels, Why quantum? Strategic, scientific, and technological significance, A snapshot of quantum technologies: Computing, Communication, and Sensing, National and global quantum missions: India's Quantum Mission, EU, USA, China

Introduction

Quantum theory, also known as **quantum mechanics**, is a fundamental theory in physics that describes the behavior of matter and energy at the smallest scales — atoms and subatomic particles. Unlike classical physics, quantum theory introduces concepts that defy everyday experience, such as **superposition**, **entanglement**, and **quantum uncertainty**.

At its core, quantum theory explains how particles like electrons and photons exist not just as particles, but also as **probabilistic wavefunctions**, governed by **Schrödinger's equation**. Measurement in quantum mechanics collapses this wavefunction into a definite state — an idea that introduces randomness and observer-dependence into physics.

Quantum technologies leverage these **principles** to develop revolutionary tools and systems with applications in **computing**, **communication**, **sensing**, and more. Global interest in quantum technologies is growing rapidly, with significant investments by governments and private companies. The field is highly interdisciplinary, combining **physics**, **computer science**, **engineering**, and **mathematics**, and is poised to become a cornerstone of the next technological revolution.

1.1 The transition from classical to quantum physics:

At the end of the 19th century, classical physics was seen as nearly complete. Newtonian mechanics explained planetary motion, Maxwell's equations described **electromagnetism**, thermodynamics clarified heat and work, and classical optics enabled scientific tools like microscopes and telescopes. Chemistry had a basic periodic table, but the atomic structure and electron behaviour in reactions were not understood.

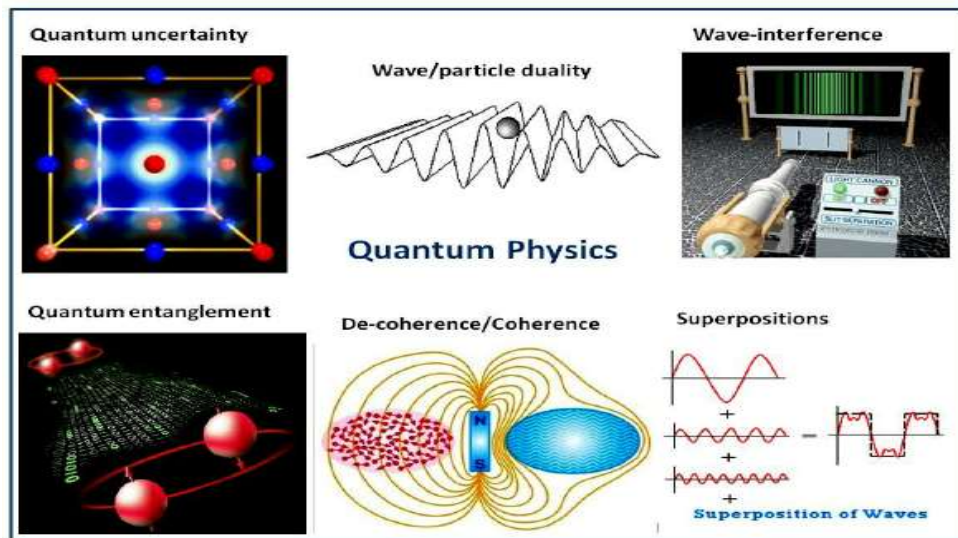
The **transition** from classical physics to quantum mechanics marks one of the most profound paradigm shifts in scientific history. Classical physics couldn't explain phenomena, such as blackbody radiation, the photoelectric effect, and hydrogen spectral lines, molecular spectroscopy.

Physics (Classical Physics)	FEATURE	Quantum Physics
Macroscopic (large-scale objects like planets and cars)	Scale	Microscopic (atoms, electrons, photons)
Deterministic (precise predictions)	Nature	Probabilistic (only probabilities of outcomes)
Newton's Laws, Electromagnetism, Thermodynamics	Key Theories	Schrödinger's Equation, Heisenberg's Uncertainty Principle, Quantum Field Theory
Not applicable	Wave-Particle Duality	Particles can behave as waves and vice versa
Objects exist in a single definite state	Superposition	Particles can exist in multiple states simultaneously
No concept of entanglement	Entanglement	Particles can be instantaneously correlated regardless of distance

These **challenges** led to the development of quantum mechanics in the early 20th century, with major contributions from scientists like Planck, Einstein, Bohr, Schrödinger, and others.

Quantum mechanics: a theory based not on certainty, but probability, where particles like electrons exhibit both wave-like and particle-like properties. **Max Planck** introduced the idea that displaying both energy quantization (in packets

called quanta) and wave-like behavior such as interference, emitted in **discrete amounts called quanta**.



Louis de Broglie suggested that matter had **wave properties**, while **Werner Heisenberg** introduced the **uncertainty principle**, redefining how we measure physical systems. **Schrödinger's wave equation** offered a new mathematical tool to describe **electron behaviour in atoms**.

1.2 Fundamental principles:

a) Superposition:

Superposition is a fundamental concept in quantum mechanics, describing the condition in which a quantum system can exist in multiple states or configurations simultaneously. Classical bits can exist in two possible states, typically labelled as "0" and "1". In contrast, because a qubit is a quantum mechanical system, it can exist in the "0" state, the "1" state, or any state that is a linear combination of 0 and 1.

Mathematically, superposition is a linear combination of "0" and "1" and can be written as:

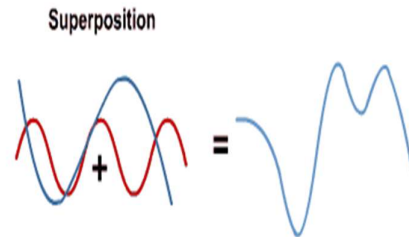
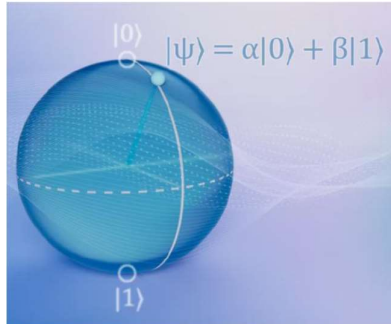
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where,

- $|\psi\rangle$ is the state of the qubit,

- $|0\rangle$ and $|1\rangle$ are the basis states (or the computational basis states), and
- α and β are complex numbers called probability amplitudes.

The probability amplitudes determine the probability of measuring the qubit in either state when a measurement is made.



Importantly, the state of superposition can be maintained only while a quantum system is unobserved. **Once measured, the wave function** of a quantum system in a state of superposition "**collapses**" into one of the basis states.

For a concrete example, of how this might work if superposition could exist in the everyday world, imagine that a **coin** that is flipped and lands on a table. In classical mechanics -- and in the everyday world as we know it -- the coin ends up in a state of either heads or tails. In a quantum mechanical system, the coin could be both heads and tails at the same time.

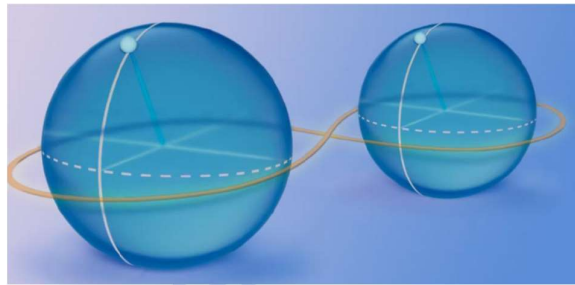
Superposition is central to **quantum computing**, offering a level of power and parallelism that classical computers cannot match. With classical bits, each bit can represent only a single value at a time, limiting operations to sequential logic. But with qubits in superposition, a quantum computer can evaluate many possible combinations of inputs simultaneously. This effect is known as **quantum parallelism**.

For example, a system of just 3 qubits can represent 8 states at once (2^3), allowing it to process all those configurations in a single operation.

Technologies like quantum sensors, quantum simulators, and quantum cryptography also leverage this principle. As researchers continue to refine hardware and error correction techniques, harnessing superposition will remain key to unlocking the full promise of quantum technologies in the years ahead.

b) Entanglement:

Entanglement is a fundamental concept of quantum mechanics that describes a non-classical **correlation**, or shared quantum state, between **two or more quantum systems** (or quantum particles) even if they are separated by a large distance. This phenomenon is also known as **quantum non-locality**, and it is one of the key features of quantum mechanics that distinguishes it from classical mechanics.



Quantum systems are described by a mathematical object called a **wavefunction**, which contains information about the possible outcomes of measurements that can be performed on the systems. When two or more quantum systems are **entangled**, their wavefunction cannot be expressed as a **product of individual wavefunctions** for each system.

The fact that **entangled** systems are described by a single wavefunction means that any actions or **measurements made on one of the systems affect the state of the other systems**.

Instead of manipulating each qubit individually, as in classical computing, **entanglement** allows quantum computers to manipulate many qubits in a single operation which enables **quantum parallelism**.

For example, consider two qubits that are initially prepared in an entangled state. If a measurement is made on one of the qubits, and it is found to be in the **state** $|0\rangle$, **then the state of the other qubit immediately collapses to the state** $|0\rangle$ **as well**. Similarly, if the first qubit is measured to be in the state $|1\rangle$, then the state of the second qubit collapses to the state $|1\rangle$ as well.

Entanglement also plays a vital role in **quantum teleportation**, which does not transmit matter but instead transfers a quantum state from one location to another using pre-shared entangled qubits and classical communication.

Entanglement is also a key resource for **quantum error correction**, which is necessary to protect quantum information from decoherence and other errors. By creating and manipulating entangled states, quantum computers can detect and correct errors without disturbing original information - which is not possible for classical computers.

c) Uncertainty Principle:

The **Uncertainty Principle**, also known as **Heisenberg's Uncertainty Principle**, is a fundamental concept in **quantum mechanics**. It states that:

“It is impossible to simultaneously know both the exact position and the exact momentum of a particle”.

$$\Rightarrow \Delta p \cdot \Delta x \geq \frac{h}{4\pi}$$

$$\Rightarrow m\Delta v \cdot \Delta x \geq \frac{h}{4\pi}$$

Where, Δp = uncertainty in momentum

Δx = uncertainty in position

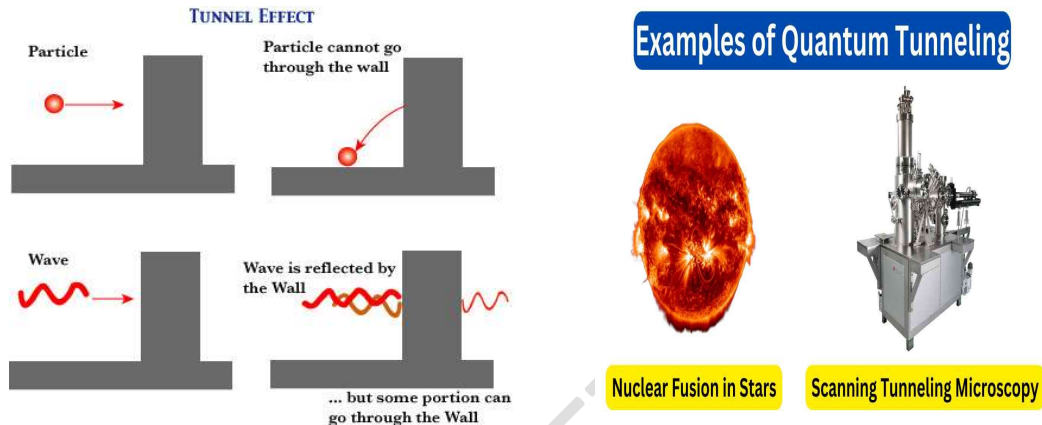
m = mass of particle

h = Planck's constant

This equation reveals that we cannot know both the location and momentum of an electron. If we know the momentum with more certainty (**smaller Δmv**), then the location is less known (**larger Δx**). The reverse is also true.



It helps explain phenomena like zero-point energy, where particles have motion even at absolute zero temperature, and **quantum tunnelling**, where particles appear to pass through energy barriers.



It is also crucial in **quantum cryptography**, helping to ensure that any eavesdropping attempt on a quantum communication channel can be detected, since measurement disturbs the system.

d) Wave-particle duality:

Wave-particle duality is a fundamental concept in quantum mechanics which reveals that all quantum objects, including light and matter, exhibit both wave-like and particle-like properties.

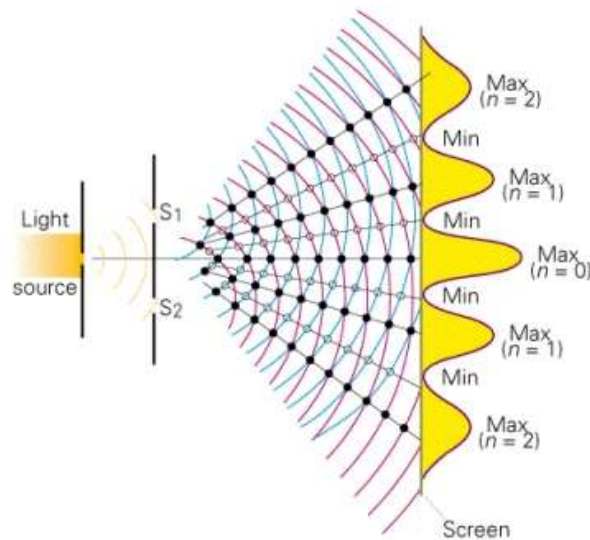
Light, which was classically understood as a wave due to its ability to interfere and diffract, was shown by Einstein in 1905 to also behave like a stream of particles called photons when explaining the photoelectric effect—where light knocks electrons out of a metal surface.

Louis de Broglie in 1924 proposed that matter, such as electrons, should also exhibit wave-like behavior. He introduced the concept of the matter wave, assigning a wavelength to any particle based on its momentum using the relation,

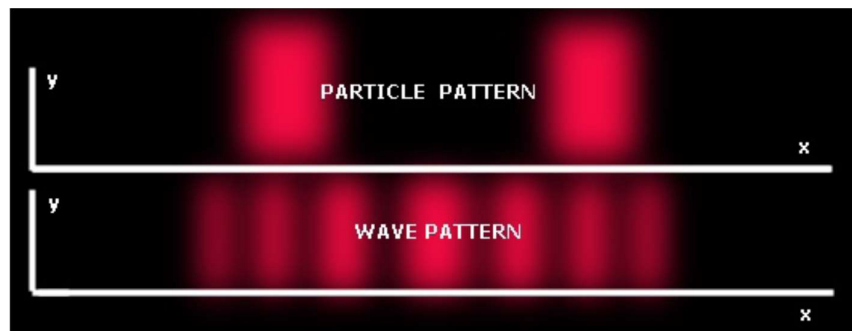
$$\lambda = \frac{h}{p},$$

Where, h is Planck's constant.

This duality was dramatically confirmed by experiments such as the electron **double-slit experiment**. When electrons pass through two slits, they produce an **interference** pattern typical of waves.



In quantum theory, particles are described by wavefunctions, which represent the probability of finding a particle in a certain state. Only when a measurement is made does this wavefunction “**collapse**” to a single, definite value. Thus, the wave-particle duality reveals that quantum entities exist in a **superposition** of possibilities until observed.

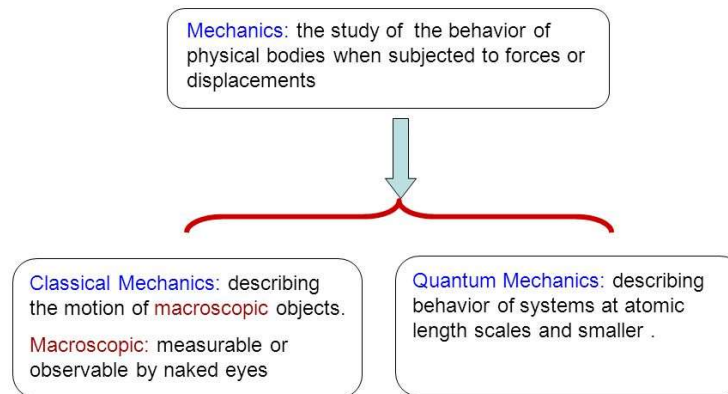


A photon beam when passed through **one slit at a time**, or observed using detectors in front of each slit, light behaves as **particle** (i.e. **no interference pattern**). **Without observation**, photon was in position of two states "**top**" and "**bottom**", going through both at same time. Observation "**collapses**" wave function of particle. This is called as **uncertainty principle**.

This concept not only underpins the foundations of quantum physics but also drives the functioning of quantum technologies like electron microscopes, quantum computers, and even lasers.

1.3 Classical vs Quantum mechanics – theoretical comparison

Classical Mechanics and Quantum Mechanics

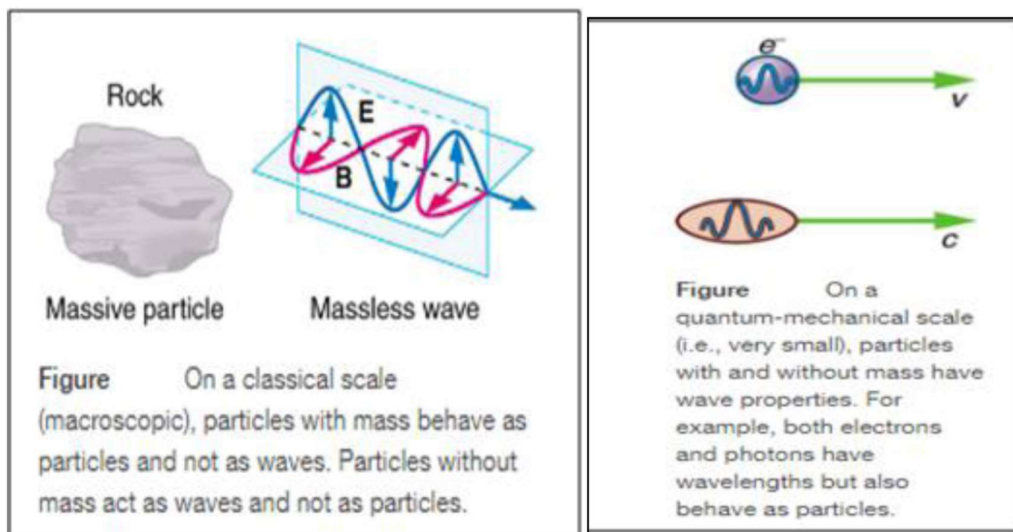


Classical mechanics and quantum mechanics are two distinct frameworks for understanding physical phenomena. **Classical mechanics**, formulated primarily by **Newton**, governs the **motion of macroscopic objects** like planets, cars, and projectiles.

Quantum mechanics, developed in the early 20th century, is essential for accurately describing the behavior of **microscopic particles** such as **electrons, atoms, and photons**. The two theories differ fundamentally in their assumptions, mathematical formalisms, and interpretations of nature.

In **classical mechanics**, objects always have **definite positions and velocities**. The state of a system can be precisely predicted using Newton's laws, and the

evolution of that system is **deterministic**: given initial conditions, the future behavior is uniquely determined.



On the other hand, **quantum mechanics** introduces inherent indeterminacy. A particle does **not have a definite position or momentum until it is measured**. Instead, it is described by a wavefunction, which encodes a **probability distribution** over all possible outcomes. The act of measurement **collapses** this wavefunction, resulting in a specific observed value. This is called as **uncertainty principle**.

Different phenomenon like **photoelectric effect**, black body radiation etc, have proved that Electromagnetic radiation has **particle nature**. **Optical phenomenon** like interference and diffraction have proved that Electromagnetic radiation has **wave nature**. We call this two-fold nature the **particle-wave duality**.

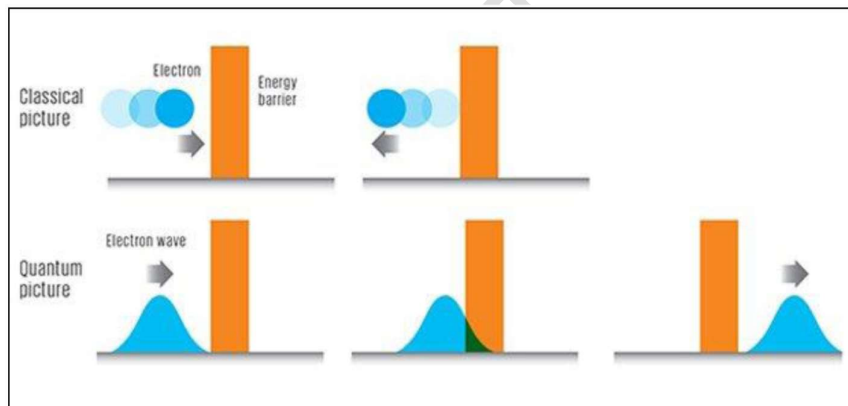
For any photon, $\nu = E/h$. The de Broglie wavelength λ , is associated with a particle and its momentum, through the Planck's constant is, $\lambda = h/p = h/mv$.

Classical mechanics relies on **continuous variables** and **smooth trajectories** in phase space. In contrast, **quantum mechanics** uses **discrete quantized energy levels** and operates within a **probabilistic framework**, governed by operators on Hilbert space and the Schrödinger equation.



Another major difference is that classical systems obey the **principle of determinism** and locality, quantum systems exhibit phenomena like **superposition** where particles exist in multiple states simultaneously, **entanglement**, and **non-locality**, where particles can exhibit strong correlations even across large distances.

According to **classical mechanics** particle with energy less than potential barrier ($E < V_0$) striking to wall of potential barrier it **cannot leak** through it.



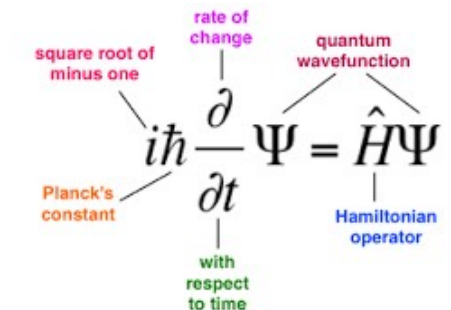
However, **quantum mechanics** principle is different than classical one and particle possess a **wave like behavior** which enables it to **leak through barrier**. This type of penetration of particles is known as **tunnelling effect**. It has important applications in the tunnel diode, quantum computing (QC), and in the scanning tunnelling microscope.

1.4 Quantum States and Measurement: Nature of Observation

In quantum mechanics, a **quantum state** represents the **complete information about a system** and is typically described by a mathematical function called a wavefunction (**denoted by Ψ**).

a) Quantum State:

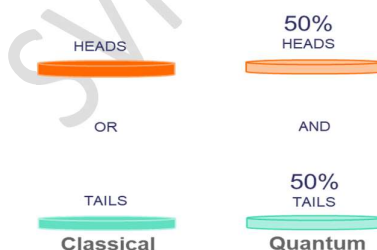
In quantum mechanics, a **quantum state** is a mathematical description of quantum system (like an electron or photon) properties at a given time and is defined by a **Schrödinger wave function (ψ)**,



$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$$

It uses a wave function, ψ , to represent the probability amplitude of a particle's position and state.

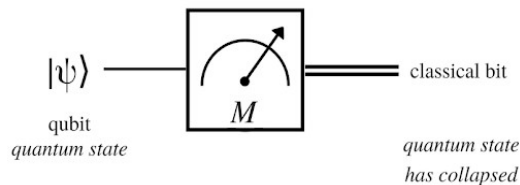
Unlike classical systems, quantum systems can exist in a **superposition** of **multiple states** simultaneously, meaning a particle can be in many possible states **until a measurement is made**. If we have a two-sided coin in the classical world, there's only **one way** to measure it: is it heads or tails?



A quantum coin, on the other hand, can be measured in infinite incompatible ways by measuring whether it's heads, tails, or a superposition of both. This phenomenon is known as **the uncertainty principle**.

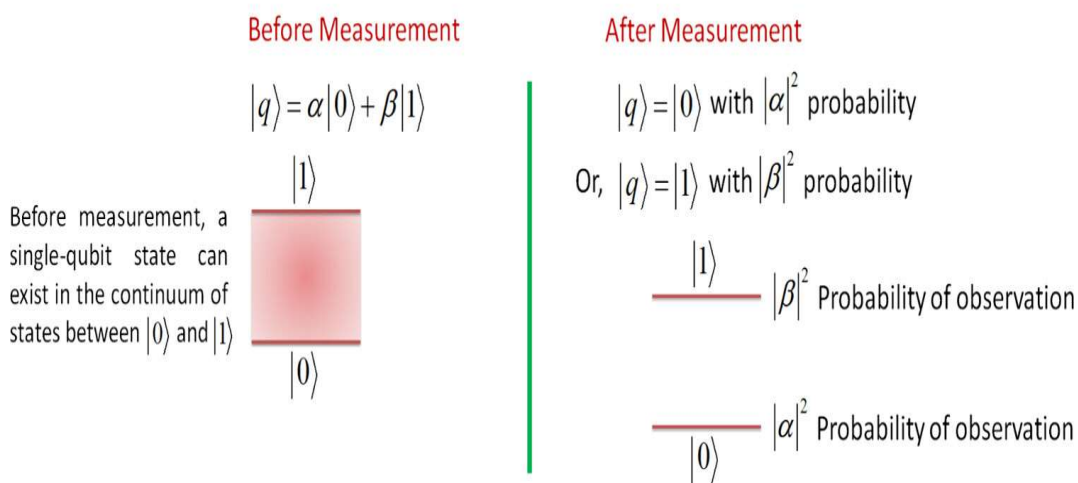
b) Collapse of the Wavefunction: (Quantum Measurement by Nature of observation)

When we **observe or measure** a quantum system, the wave function **collapses** to **one** of the possible **basis states**, and we get a definite outcome. You'll get either $|0\rangle$ with probability $|\alpha|^2$, or $|1\rangle$ with $|\beta|^2$.



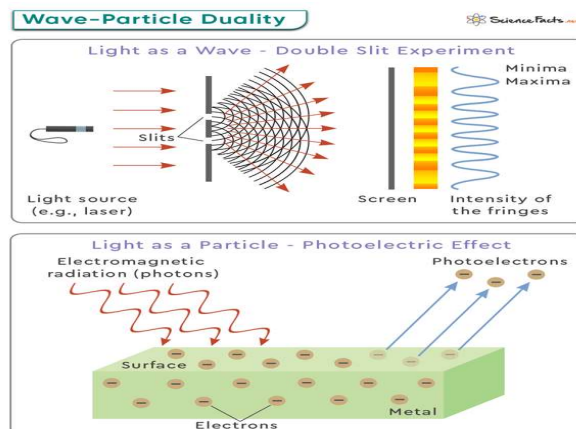
After measuring an observable **A** the system **collapses** to one of its basis states ($|0\rangle$ or $|1\rangle$). Measurement outcomes are **probabilistic**.

The Nature of Quantum Measurement



1.5 Overview of Quantum Systems: Electrons, Photons, Atoms

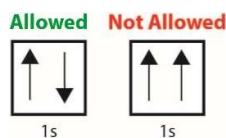
Quantum systems consist of microscopic entities such as electrons, photons, and atoms, which all exhibit wave-particle duality and are governed by the laws of quantum mechanics.



- a) **Electrons**, though traditionally thought of as point particles, also behave like waves. This wave nature is responsible for phenomena like electron diffraction and atomic orbitals.

In quantum systems, **electrons** are fundamental particles that are considered **fermions**, meaning they have **half-integer spin values** ($1/2$, $3/2$, $5/2$, etc.) and must adhere to the **Pauli Exclusion Principle**, which states that:

“No two electrons can occupy the same quantum state within a system”.



Electrons exhibit wave-particle duality:

Electrons display **particle-like** behavior in experiments such as electron diffraction, where they act as tiny particles. When electrons pass through a **double slit**, they create an interference pattern, like how **waves** behave when passing through two openings.

- b) **Photons**, on the other hand, are the massless quantum particles of light and always move at the **speed of light** and carry energy proportional to their frequency, as described by:

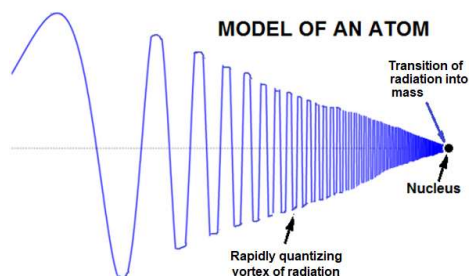
- $E = h\nu$
- $v = c/\lambda$ where, c is the speed of light.

Photons, display both **energy quantization** (in packets called quanta) and **wave-like behavior** such as interference. Their **wave-particle duality** manifests in phenomena like **interference and the photoelectric effect**. Photons can also become entangled, making them important in quantum communication and cryptography.

Photons, the quantum of light, are **bosons with integer spin values of 0,1,2, etc.** This means they are particles that can occupy the same quantum state.

- c) **Atoms** are more complex quantum systems made of a **nucleus** (protons and neutrons) surrounded by electrons in discrete energy levels. Can absorb or emit photons to **transition** between energy levels (basis of lasers, spectroscopy).

Atomic energy levels are **quantized**, forming the foundation of atomic clocks and quantum computing (with trapped ions or neutral atoms as qubits).



Neutrons, which are subatomic particles found in the nucleus of an atom, exhibit **dual behavior**. They can be diffracted by crystals, demonstrating their wave-like nature, but they also interact with matter as particles.

Each of these systems—electrons, photons, and atoms—exhibit **quantum behaviors** including superposition (being in multiple states at once), entanglement (non-local correlation between particles), and quantum tunneling (the ability to cross classically forbidden barriers).

Innovations such as semiconductors, quantum sensors, lasers, MRI machines, LEDs, and quantum computers all rely on manipulating the **quantum properties** of these particles.

1.6 The Concept of Quantization:

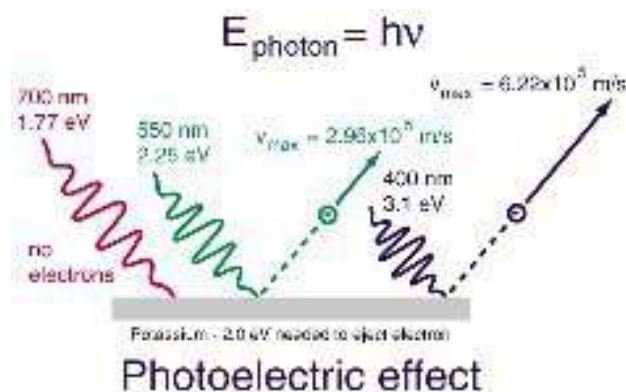
Quantization is a core principle of quantum mechanics that states certain physical quantities, like energy, angular momentum, and charge, can only take on **discrete values**. The earliest evidence for quantization came from the **blackbody radiation** problem and the **photoelectric effect**, which were explained by assuming that energy is emitted or absorbed in discrete units called **quanta**.

Blackbody radiation & Photo electric effect: Origin

Quantization was first introduced to address failures in classical theories when explaining microscopic phenomena. A **black body** is an idealised object which absorbs and emits thermal radiation based only on its temperature not its material or shape. The **spectrum** of radiation emitted by a blackbody is called **blackbody radiation**.

Classical physics predicted that a blackbody should emit **infinite energy**, which is clearly **unphysical** (as ultraviolet has shorter wavelength).

Max Planck (1900) resolved this by proposing that energy is **emitted and absorbed in discrete packets** called **quanta**. Introduced the concept of quantization, where energy levels are not continuous but discrete.

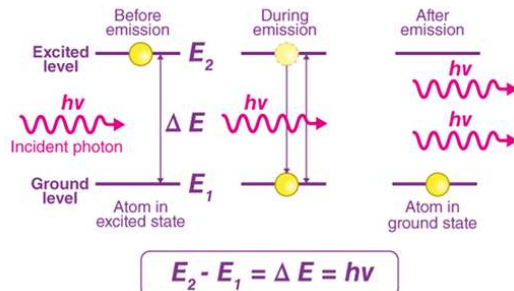


Where,

- **E** is the energy,
- **h** is **Planck's constant**, and
- **ν** is the frequency of radiation.

Albert Einstein (1905) expanded on Planck's idea to explain how light of certain frequencies can eject electrons from metal surfaces (**Photoelectric Effect**). He proposed that light consists of **photons**, each carrying energy **$E=h\nu$** confirming the **quantized nature of light energy**.

In atoms, electrons can only exist in specific **quantized energy levels**, and when an electron jumps between these levels, it **absorbs or emits energy** in quantized amounts (photons) with fixed frequencies.

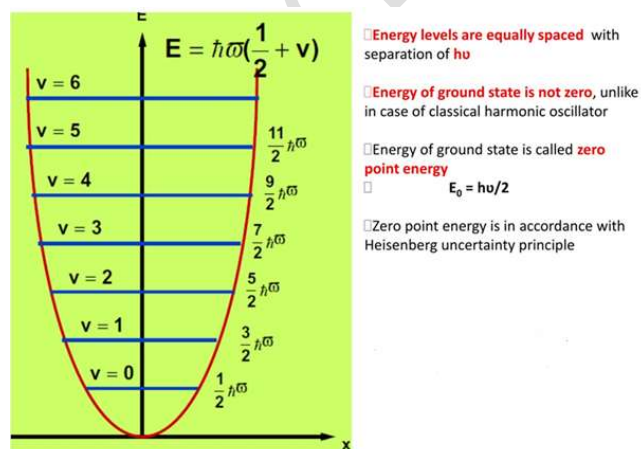


Quantization is also seen in systems like the **harmonic oscillator**, where energy levels are separated by fixed intervals. This discrete nature of quantum systems is **mathematically expressed** using **operators** with **eigenvalues** corresponding to observable quantities.

The energy eigenvalues are:

$$E_n = (v + 1/2) h\omega$$

Where, $v = 0, 1, 2, 3, \dots$. Notice once again that the energy levels are quantized. In this case they are **evenly spaced** by an amount $\Delta E = h\omega$.



In atomic systems, **quantization** becomes especially evident. Electrons in atoms cannot occupy arbitrary energy levels; instead, they are found only in certain allowed states. When an electron transitions between these levels, it absorbs or emits a photon with a specific frequency, giving rise to the spectral lines seen in emission and absorption spectra.

This phenomenon is responsible for the stability of atoms and the unique identity of elements. Similar principles apply to **rotational and vibrational states of molecules**, which are also **quantized** and form the basis of various spectroscopic techniques.

Angular momentum, spin, and even electric charge can also be quantized, leading to surprising consequences in both microscopic systems and macroscopic quantum phenomena like superconductivity and quantum Hall effects.

1.7 Why Quantum?

Traditional classical theories fail to explain the behavior of microscopic particles like electrons, photons, and atoms. Quantum theory fills this gap by accurately describing the probabilistic and non-deterministic nature of such particles.

a) Strategic Significance:

The strategic value of quantum technologies is increasingly recognized by governments and industries worldwide. Quantum computing, for instance, has the potential to break classical encryption methods, posing both risks and opportunities for national security.

Quantum communication promises ultra-secure information transfer using Quantum Key Distribution (QKD), which is theoretically unbreakable.

Nations are investing heavily in quantum research to ensure leadership in this frontier domain, as it offers a significant edge in defence, surveillance, intelligence, and cybersecurity. As a result, mastery of quantum technologies is becoming a critical component of geopolitical power and technological sovereignty.

b) Scientific Significance:

Quantum mechanics stands as one of the most profound scientific achievements of the 20th century. It explains phenomena that classical physics cannot, such as superconductivity, quantum tunneling, and the behavior of particles in extreme conditions.

Quantum theory has also laid the groundwork for fields like quantum chemistry. It enables the modeling of complex systems with high accuracy, leading to discoveries in materials science, nanotechnology, and fundamental physics.

Quantum theory allows scientists to model atomic and subatomic systems with remarkable precision, leading to the discovery of new materials and deeper insights into the behavior of matter and energy.

c) Technological Significance:

Quantum technologies are set to redefine the future of computation, communication, sensing, and imaging.

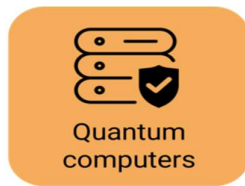
Quantum computers can solve certain classes of problems exponentially faster than classical computers, with potential applications in drug discovery, optimization, machine learning, climate modelling, and logistics optimization.

Quantum sensors can measure time, magnetic fields, and gravitational forces with unprecedented precision, useful in GPS systems, medical diagnostics, and geological surveys.

Meanwhile, quantum cryptography offers solutions for secure digital infrastructure.

1.8 A Snapshot of Quantum Technologies: Computing, Communication, and Sensing:

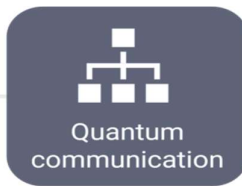
Quantum technologies are at the forefront of a technological revolution, harnessing the unique and counterintuitive principles of quantum mechanics—such as superposition, entanglement, and quantization.



Quantum computers

Calculations using quantum bits can solve certain problems faster. Used for:

- Physical simulation
- Optimisation
- Machine learning
- Cryptography



Quantum communication

Interception of quantum information can be detected. Used for:

- Quantum key distribution
- Quantum internet
- Blind quantum computation



Quantum sensors

Extremely sensitive to physical forces. Used for:

- Timing
- Navigation
- Random numbers
- Antennas
- Industrial sensors

These technologies are being developed across three primary domains: **quantum computing**, **quantum communication**, and **quantum sensing**, each offering transformative potential for science, industry, and society everyday life.

Quantum Computing is perhaps the most well-known application of quantum mechanics. Unlike classical computers that process information in binary bits (0 or 1), quantum computers use quantum bits or *qubits*, which can exist in superpositions of 0 and 1.

This allows quantum computers to perform many calculations in parallel, enabling them to solve certain problems—like factoring large numbers, simulating molecular behavior, or optimizing complex systems—exponentially faster than classical machines.

Below is an image of the **interior** of an **IBM Quantum System One** quantum computer, one of approximately 20 that form the core of IBM's quantum computing efforts.

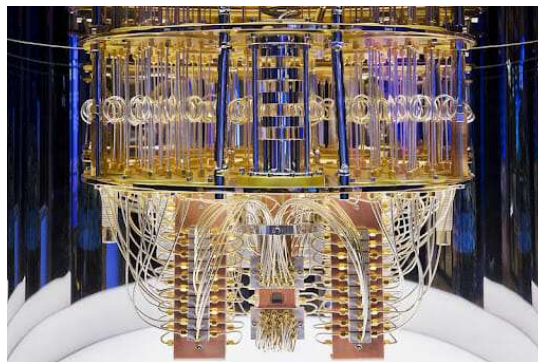
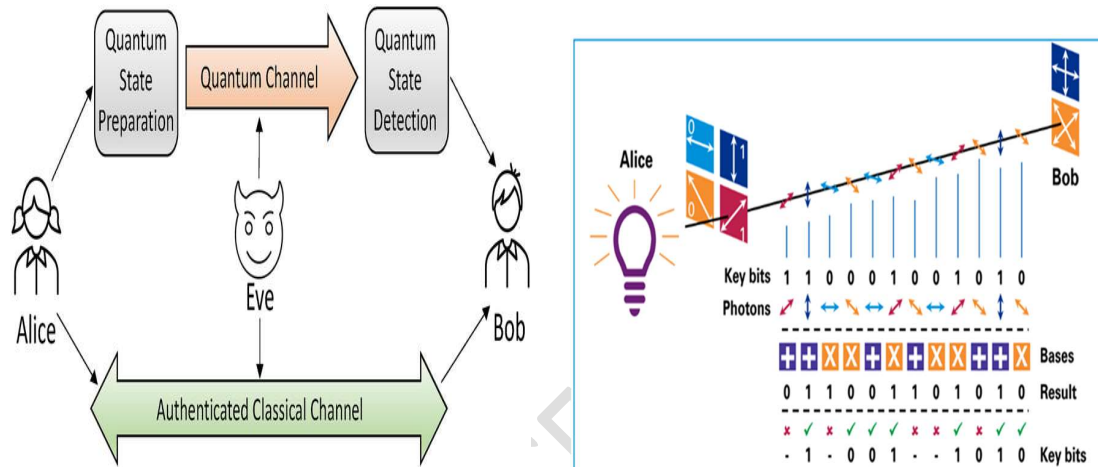
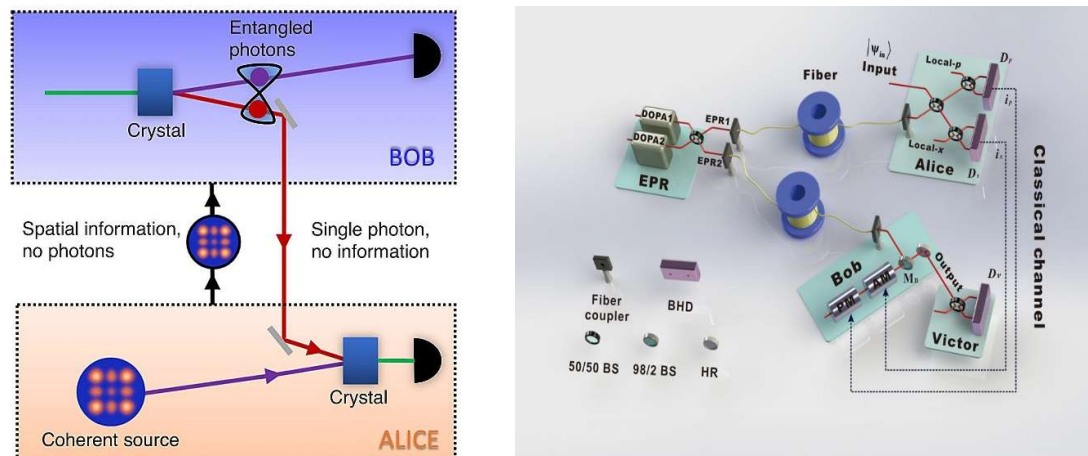


Fig: IBM Quantum System One (Internal View)

Quantum Communication focuses on the secure transmission of information using quantum phenomena. The most notable technique is **Quantum Key Distribution (QKD)**, which allows two parties to share a cryptographic key with security guaranteed by the laws of physics. If an eavesdropper tries to intercept the key, the quantum state is disturbed, alerting the communicating parties.



Quantum communication can also involve **quantum teleportation**, where the state of a particle is transferred instantaneously over a distance, using entanglement.



In **Quantum Sensing**, quantum sensors can detect minute variations in gravitational fields, magnetic fields, acceleration, and time. Applications range from

medical diagnostics (e.g., highly sensitive brain scans using quantum magnetometers), to underground exploration, to navigation systems that do not rely on **GPS**. **Atomic clocks**—among the most accurate devices ever built—are based on quantum transitions and are crucial **for global positioning and timekeeping systems**.

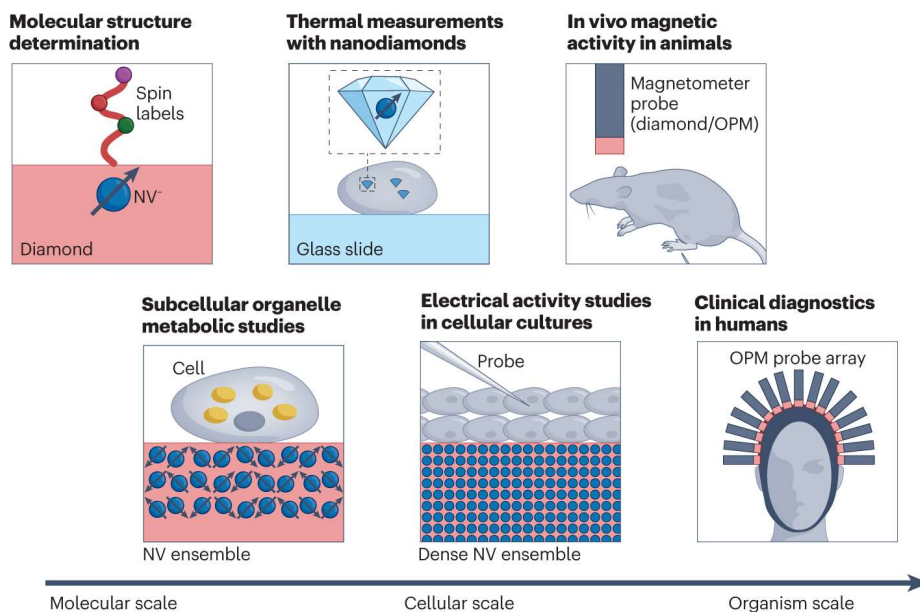


Fig: Quantum sensors for Bio-Medical Applications

1.9 National and Global Quantum Missions:

As quantum technologies emerge as a critical area of innovation and national interest, several countries have launched ambitious quantum missions to secure strategic and technological leadership.

India, along with major global powers like the USA, China, and the European Union, is actively building its presence in the quantum landscape.

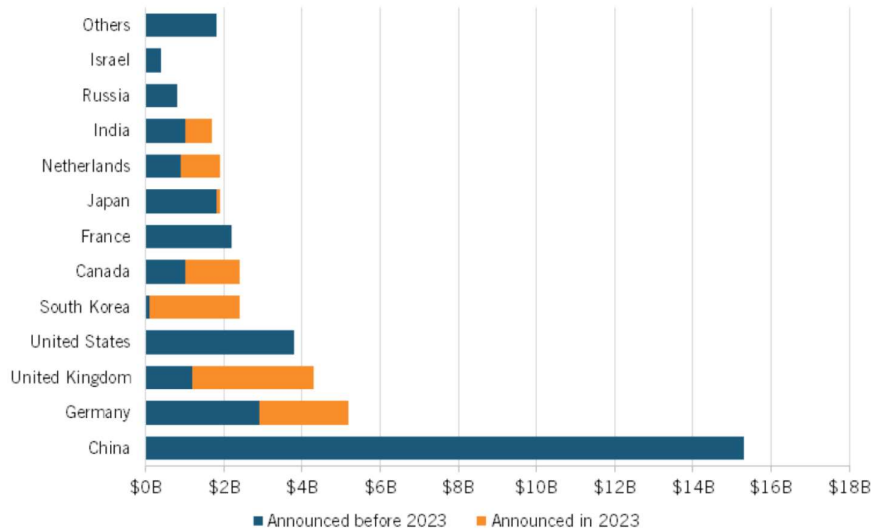


Fig: Announced government investments in quantum research and commercialization around the world.

China leads in quantum communications, lags behind in computing (where the United States excels), and matches the United States in sensing, excelling in market-ready tech, while the United States dominates high-impact areas.

a) India's National Quantum Mission (NQM):

India launched its **National Quantum Mission** in 2023, with a budget of **₹6,003 crore (approximately \$730 million)** over eight years. The mission seeks to position India among the top quantum nations by developing indigenous capabilities in quantum computing, quantum communication, quantum sensing, and quantum materials.

It aims to establish four **Thematic Hubs (T-Hubs)** in leading academic and research institutions focusing on foundational technologies. The NQM also plans to build intermediate-scale quantum computers (with 50–1000 qubits), develop quantum key distribution networks, and promote workforce training and international collaborations.

The mission aligns with India's larger vision of self-reliance in strategic technologies and aims to boost national security, telecommunications, and advanced research.

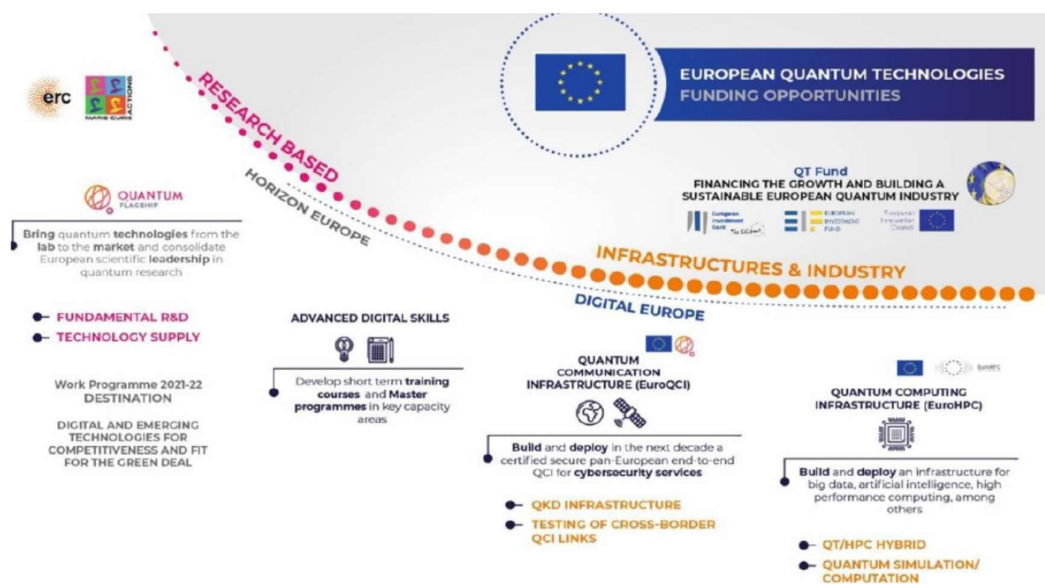


Fig: India Approves National Quantum Mission

b) European Union (EU) Quantum Flagship

The **EU Quantum Flagship** is a €1 billion, 10-year initiative launched in 2018 to unify Europe's fragmented quantum research landscape. It supports hundreds of research institutions, startups, and industries across member states.

The program focuses on four main areas: **quantum communication**, **quantum simulation**, **quantum computing**, and **quantum metrology**. The EU also promotes infrastructure projects like the **European Quantum Communication Infrastructure (EuroQCI)**, which aims to establish a secure pan-European quantum communication network.



c) United States: National Quantum Initiative Act:

The **United States** formalized its quantum strategy with the **National Quantum Initiative Act** passed in 2018.

This act coordinates efforts across government agencies, including the Department of Energy (DOE), National Science Foundation (NSF), and National Institute of Standards and Technology (NIST), with significant funding and collaboration with private sector leaders like IBM, Google, and Microsoft.



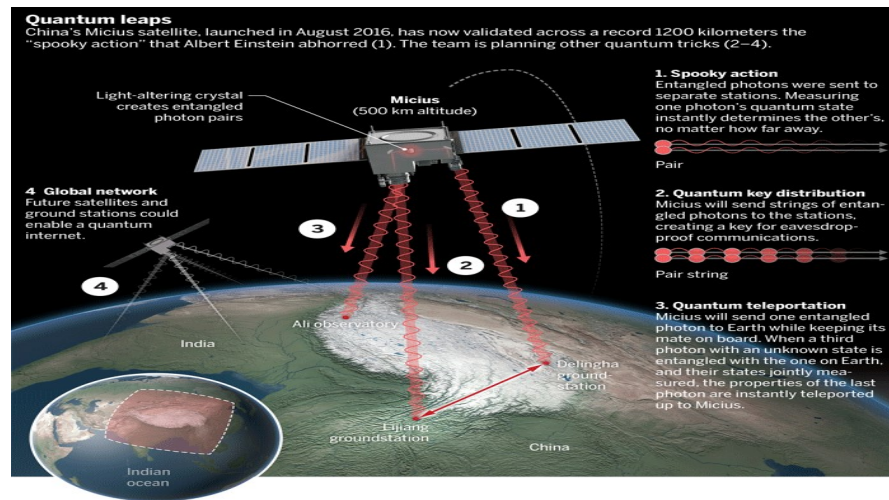
Fig: USA must dominate quantum technology

The **National Quantum Coordination Office** oversees these efforts, focusing on quantum research, technology transfer, education, and the creation of quantum research centres.

The U.S. aims to maintain its technological leadership, secure supply chains, and harness quantum advantages for national security, scientific progress, and economic growth.

d) China: Quantum Strategic Advantage

China has emerged as a global leader in quantum technology through sustained state-led investment and rapid deployment. It has achieved several milestones, including launching the world's first **quantum communication satellite (Micius)** and demonstrating **satellite-based quantum key distribution** over thousands of kilometers



China also leads in building a nationwide **quantum communication backbone network**, connecting major cities through ultra-secure fibre links. The Chinese government has reportedly invested billions of dollars in **quantum R&D**, and projects like the **National Laboratory for Quantum Information Science in Hefei** aim to consolidate China's dominance in this space. China views quantum technologies as essential to future economic and military competitiveness.

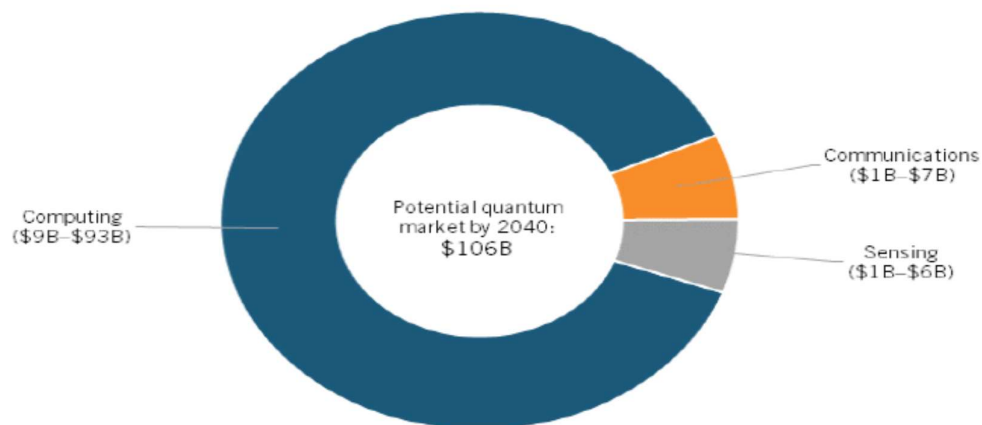


Fig: How Innovative Is China in Quantum?

Review Questions:

1. Explain in detail about transition from classical physics to quantum physics.
2. Describe the fundamental principles of quantum mechanics - Superposition, Entanglement, Uncertainty Principle, and Wave-Particle Duality.
3. Distinguish classical mechanics and quantum mechanics.
4. Discuss the concept of quantum states and the nature of quantum measurement.
5. Explain quantization and give examples of discrete energy levels in atoms, photons, and electrons.
6. Discuss major quantum technologies (**computing, communication, and sensing**) and mention important national and global quantum missions.

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